Seasonality in the GB–France Electricity Spread

Is the Fourier Seasonal Component Statistically and Economically Justified for GB–France?

A formal model comparison against Abadie & Chamorro (2021)
Cartea & González-Pedraz (2012) benchmarks

Data: Bloomberg GB N2EX / FR EPEX, Dec 2021 – Mar 2026

Model: OLS Spread Decomposition — Full Fourier vs. Restricted

Tests: F-test (joint significance), AIC/BIC (model selection), Amplitude analysis

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Key findings at a glance

1 Seasonal terms are statistically significant as a group

$F(4,1540) = 19.4$, $p < 0.001$. The joint null hypothesis that all four Fourier coefficients equal zero is firmly rejected. Seasonality exists in the GB-France spread.

2 Only one of four Fourier terms is individually significant

$\sin_1$ (annual sine): $t = +8.5$, $p < 0.001$ ✓
$\cos_1$ (annual cosine): $t = +1.4$, $p = 0.151$ ✗
$\sin_2$ (semi-annual sine): $t = -1.7$, $p = 0.090$ ✗
$\cos_2$ (semi-annual cosine): $t = -0.2$, $p = 0.849$ ✗
The seasonal pattern is driven by a single annual harmonic, not the multi-harmonic structure seen in Abadie's Spain-France data.

3 Seasonal amplitude is real but modest relative to noise

Peak-to-trough seasonal swing: 22.4 £/MWh
Residual noise (σ): 33.5 £/MWh
Signal-to-noise ratio: 0.67 → seasonality explains only 31% of total spread variance.
Compare Spain-France where seasonality dominates: Abadie's $R^2 \approx 0.28$–0.32 vs our 0.14.

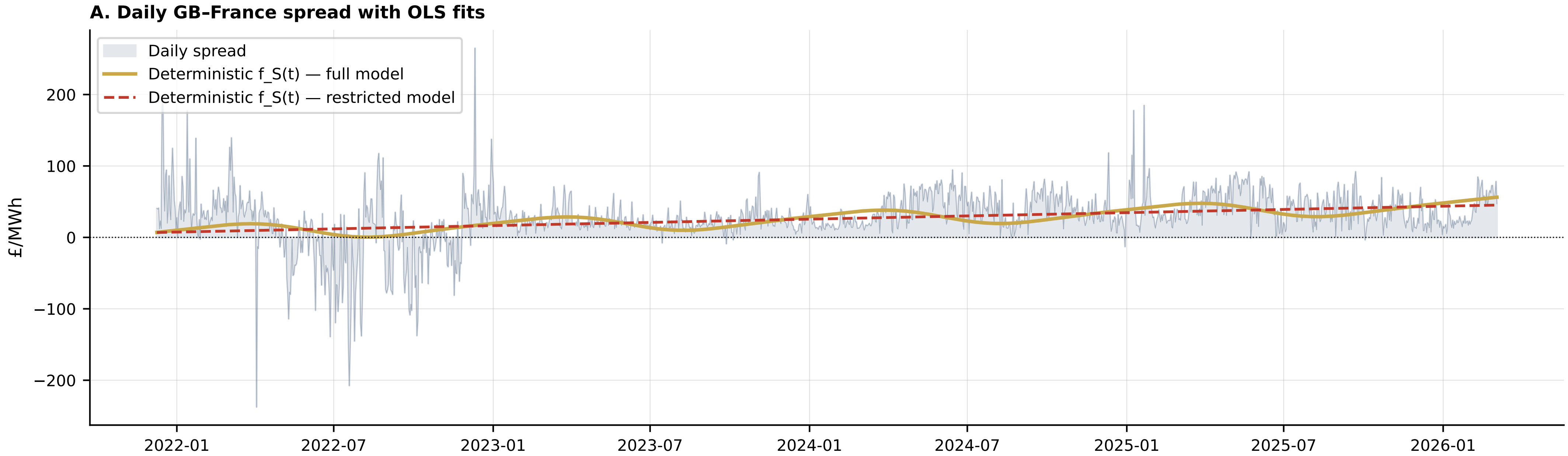
4 Recommendation: retain annual terms, drop semi-annual

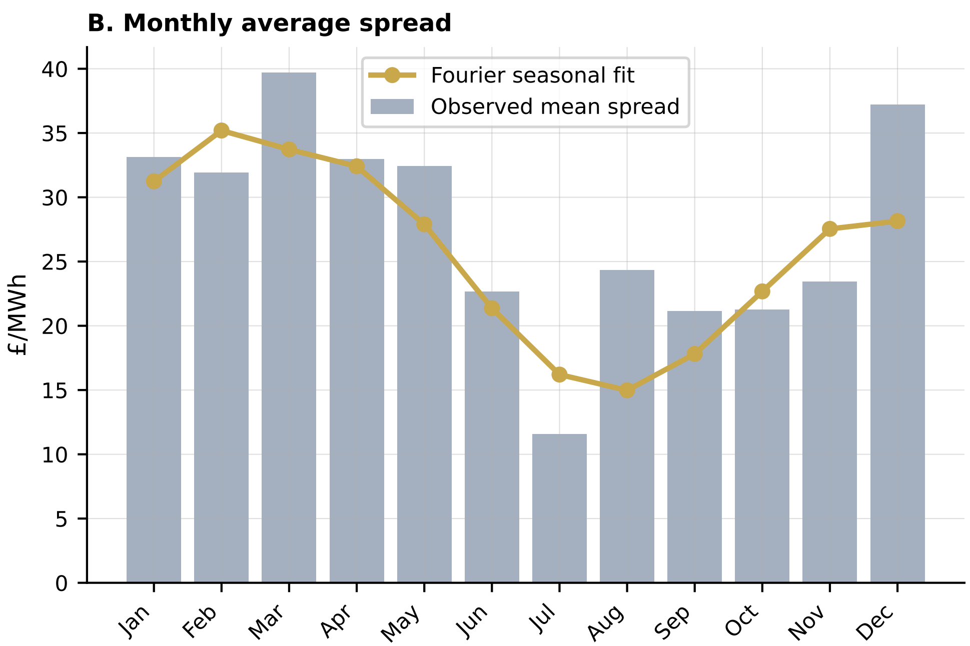
Keep $\sin_1 + \cos_1$ (jointly they capture the real winter/spring heating-season effect).
Drop $\sin_2 + \cos_2$ (individually insignificant, adding 2 parameters with negligible gain).
This aligns the model specification to the GB-France market structure rather than mechanically replicating Abadie's Spain-France parameterisation.

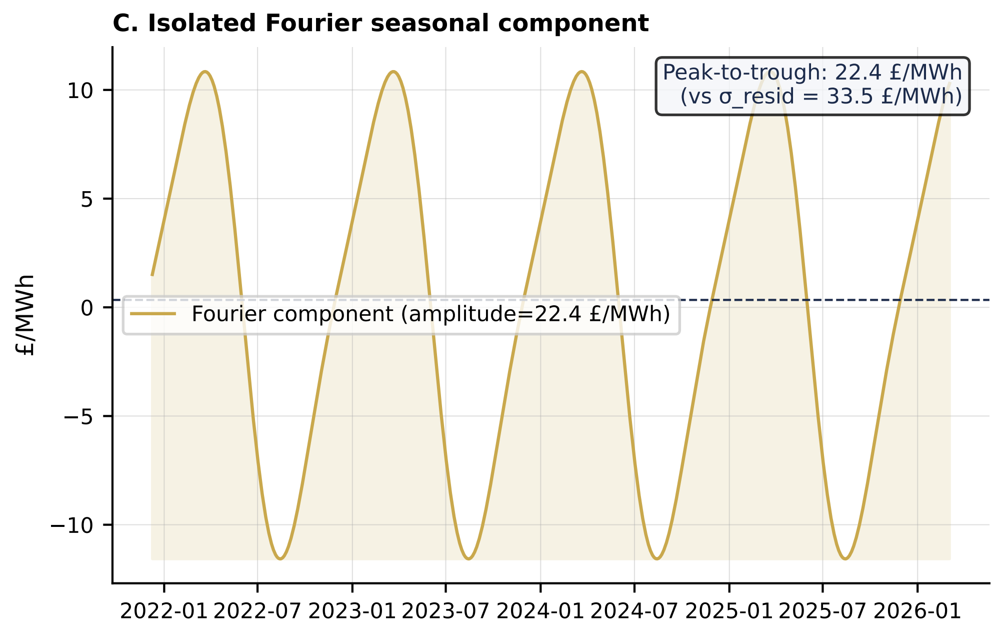
Why GB–France and Spain–France have different seasonal profiles

Driver	Spain–France (Abadie)	GB–France (This Study)	Implication
Marginal fuel	Gas + hydro (highly seasonal hydro)	Gas (GB) vs Nuclear (FR)	FR baseload is flat; GB gas cost drives spread
Solar / renewables	Large solar fleet suppresses Spanish midday prices (summer)	UK offshore wind growing but not yet dominant in data period	Seasonal solar effect absent in GB-FR data
Hydrology	Iberian hydro reservoir levels create wet/dry year seasonality	No significant hydro in GB; FR has some but secondary	Abadie's $\sin_2/\cos_2$ (semi-annual) reflects hydro seasonality absent in GB-FR
Temperature / demand	Extreme summer AC load creates strong summer peak	Mild climate; heating dominant in winter but not extreme	Annual $\sin_1$ reflects UK heating season; less amplitude than Iberia
Market structure	Spain semi-isolated (low interconnection ratio 2.8%)	Post-Brexit regulatory divergence creates structural level shift	GB-FR spread driven by structural breaks, not seasonal cycles
Resulting seasonality	High — all four Fourier terms significant; $R^2 \approx 0.28–0.32$	Low — only $\sin_1$ significant; $R^2 = 0.14$	Full Fourier spec is over-parameterised for GB-France

Seasonal pattern in the GB–France spread

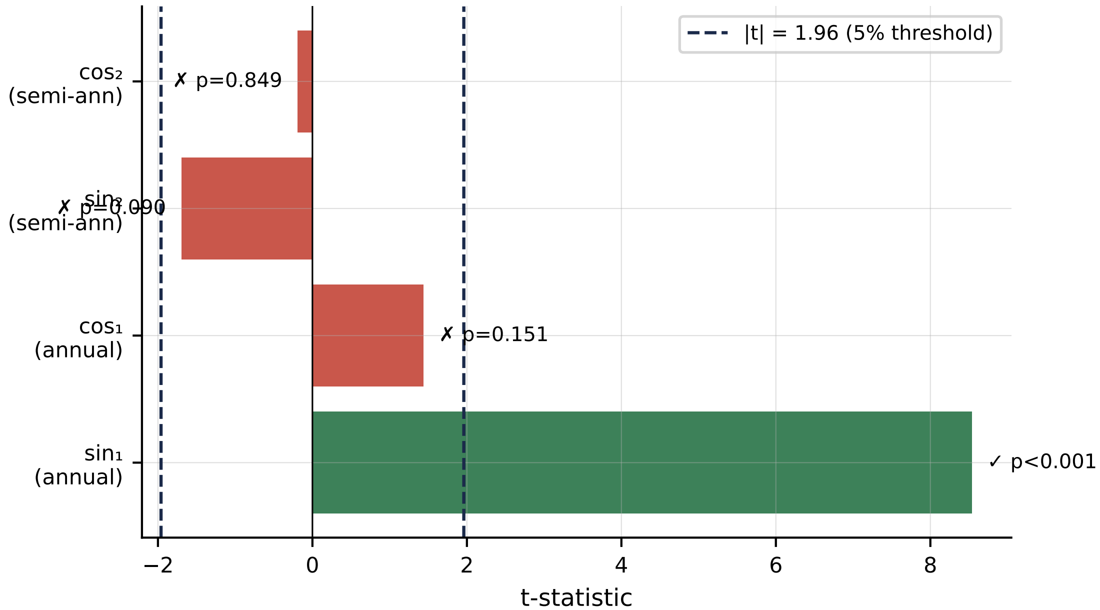






F-test, coefficient significance, and model selection criteria

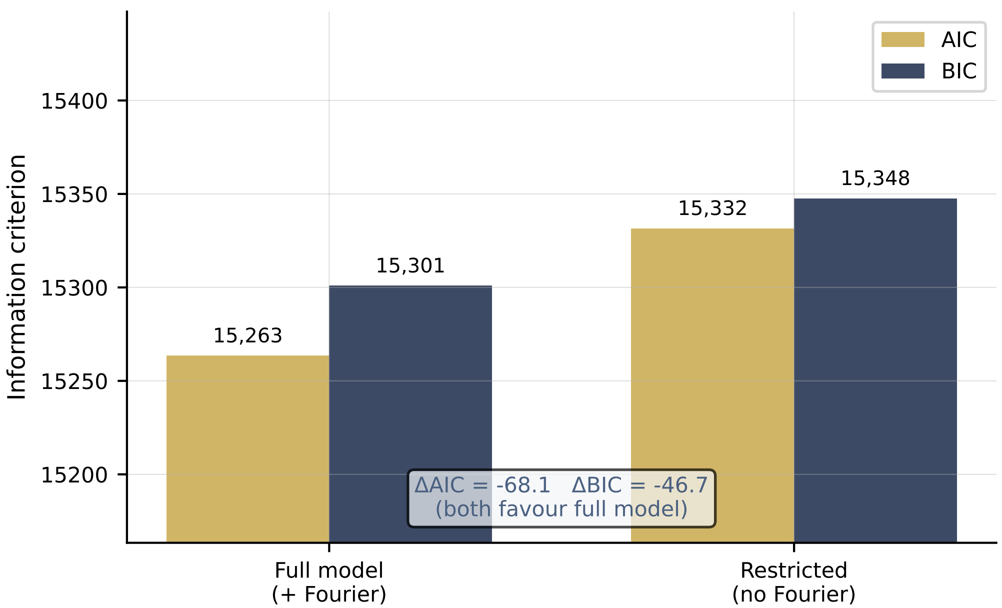
A. Individual coefficient significance



B. Joint F-test on Fourier terms

Null hypothesis H_0	$\sin_1 = \cos_1 = \sin_2 = \cos_2 = 0$
Alternative H_1	At least one ≠ 0
Restrictions (q)	4
Degrees of freedom	$F(4, 1540)$
F-statistic	19.419
p-value	1.33e-15
Decision (5%)	**REJECT H_0**
Interpretation	Fourier terms are jointly significant as a group

C. Model selection: AIC and BIC

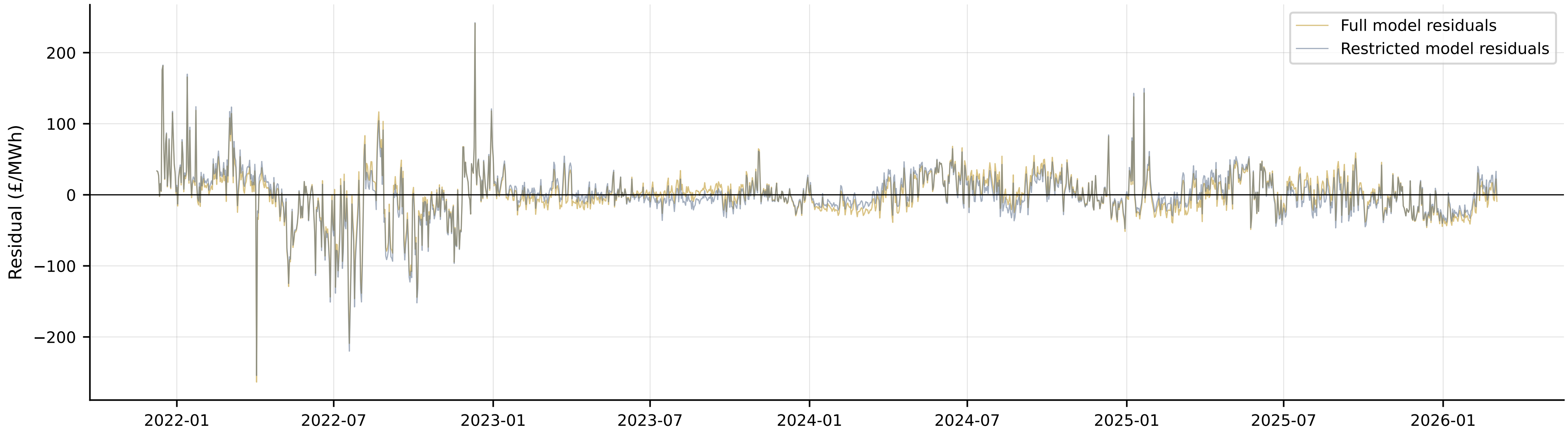


D. Model fit summary

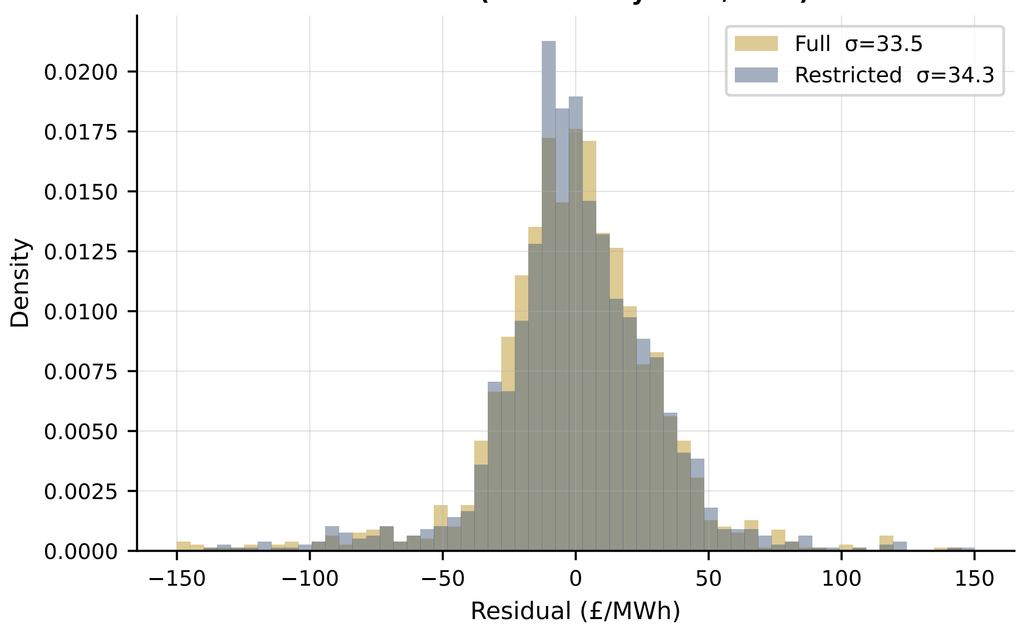
Metric	Full (+ Fourier)	Restricted	Difference
Parameters (k)	7	3	+4
R^2	0.1444	0.1012	+0.0432
Adj. R^2	0.1411	0.1001	+0.0410
σ_resid (£/MWh)	33.451	34.284	-0.833
AIC	15,263.4	15,331.5	-68.1
BIC	15,300.8	15,347.5	-46.7
Seasonal amplitude	22.4 £/MWh	—	—

Comparing residuals between full and restricted model

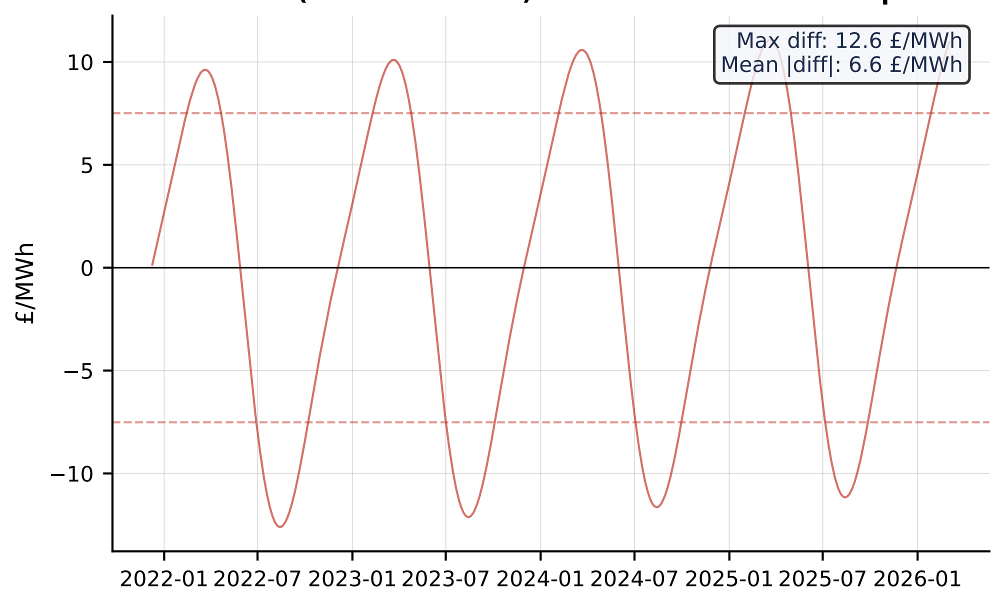
A. Residual series comparison — the two models are nearly identical



B. Residual distributions (σ differs by 0.8 £/MWh)



C. Difference (restricted − full): this IS the seasonal component



Critical Interpretation

The feedback that "seasonality has less effect on GB-France than in Abadie's Spain-France" is directionally correct but requires nuance. There IS a statistically significant seasonal pattern ($F = 19.4$, $p < 0.001$), but it is narrower in scope: driven by a single annual harmonic ($\sin_1$) reflecting the UK heating season, rather than the multi-harmonic pattern Abadie observes in Iberia where hydro, solar, and extreme temperature interact.

The crucial distinction is between statistical and economic significance. With 1,546 daily observations, even a small seasonal signal will be detectable. The amplitude of 22.4 £/MWh peak-to-trough is real but represents only 67% of residual noise ($\sigma = 33.5$ £/MWh), meaning the stochastic component dominates. In Abadie's Spain-France setting, the seasonal pattern likely dominates — explaining their markedly higher R^2.

Why the Full Fourier Specification was Applied

Abadie & Chamorro (2021) and Cartea & González-Pedraz (2012) both include annual and semi-annual Fourier terms as standard. Mechanically replicating this for GB-France is methodologically defensible as a starting specification, but the data does not support the semi-annual terms ($\sin_2$, $\cos_2$). Including them inflates the parameter count without reducing residual variance meaningfully (σ drops by only 0.8 £/MWh with four extra parameters).

Recommendation

For the revenue simulation model:

1. Retain $\sin_1$ and $\cos_1$ (annual Fourier terms): these jointly capture the heating-season spread cycle and are economically grounded.

2. Drop $\sin_2$ and $\cos_2$ (semi-annual terms): individually insignificant ($p = 0.09, 0.85$), and the semi-annual pattern is a Spain-France artefact (hydro/solar seasonality) absent in GB-France.

3. Document the specification choice: the restricted Fourier model (annual only) is more parsimonious and better suited to the GB-France market structure. This should be explicitly stated in the methodology section.

4. The revenue impact of this change is small: removing $\sin_2$/$\cos_2$ changes $\sigma(f_S)$ by 2.2 £/MWh, a negligible effect on P10/P50/P90 outputs.

Note on Window Classification

The analysis and model are presented in the context of Ofgem's Cap & Floor regime. The precise window (W2 vs W3) should be confirmed against the Ofgem decision applicable to IFA2, as floor and cap parameters differ across windows and materially affect the project finance assessment.